

CSE251: ASSIGNMENT-2

Full Marks: 60

1. [CO3]

- a) **Design** a circuit using OP-Amp that follows the VTC shown in Figure 01.
- b) Draw the waveform of V_O when $V_{in} = 2\sin(\omega t)$

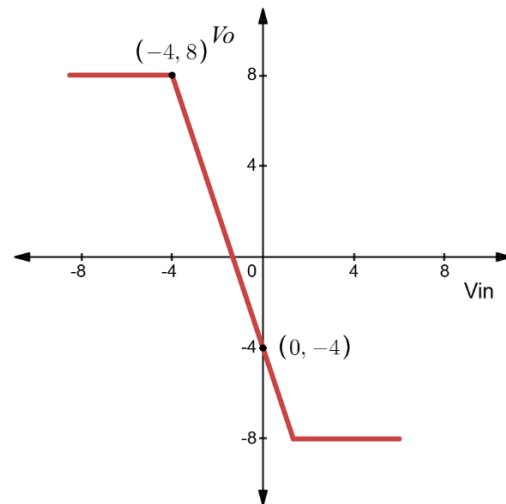
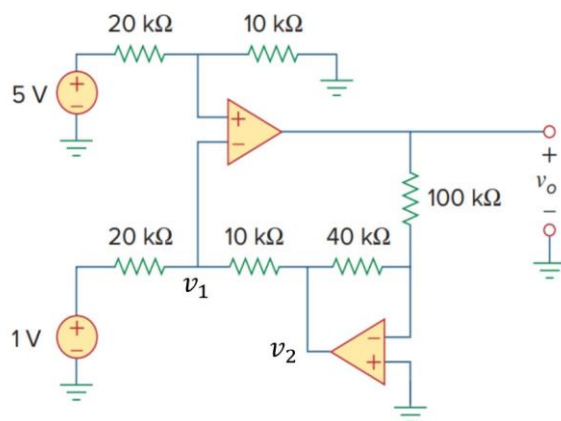


Figure 01

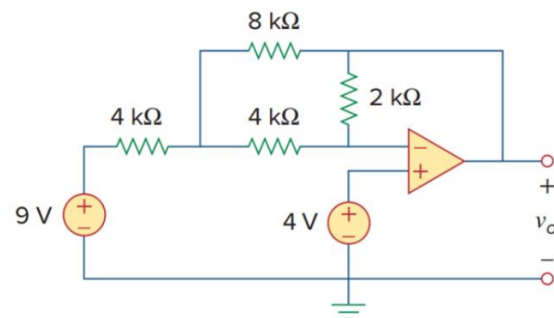
2. [CO2]

a.



Determine V_1 , V_2 and V_O for the circuit shown above.

b.



Determine V_O for the circuit shown above

3. [CO2]

Analyze the circuit to determine which diode(s) are conducting. Additionally, calculate the power dissipated by the diodes. Use $V_{D0} = 0.7\text{ V}$ for all the diodes.

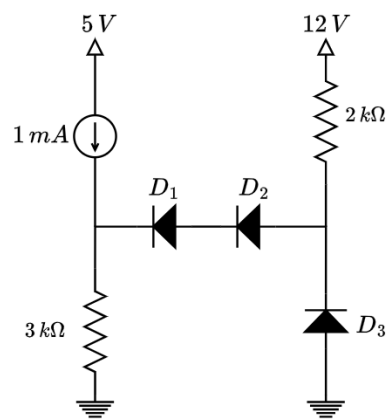


Figure 02

4. [CO2]

Analyze the circuit shown in figure-03 to Find I_{D1}

and I_{D2} for a) $R = 100\text{K}\Omega$ and

b) $R = 0.02\text{K}\Omega$

Use CVD model with $V_{D0} = 0.7\text{V}$.

[Validate assumptions]

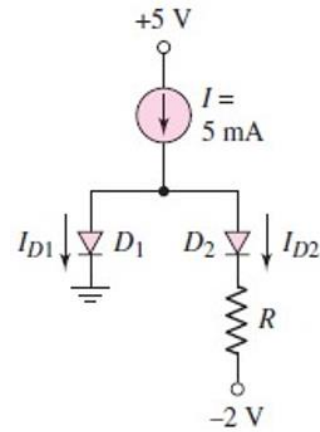


Figure 03

5. [CO2]

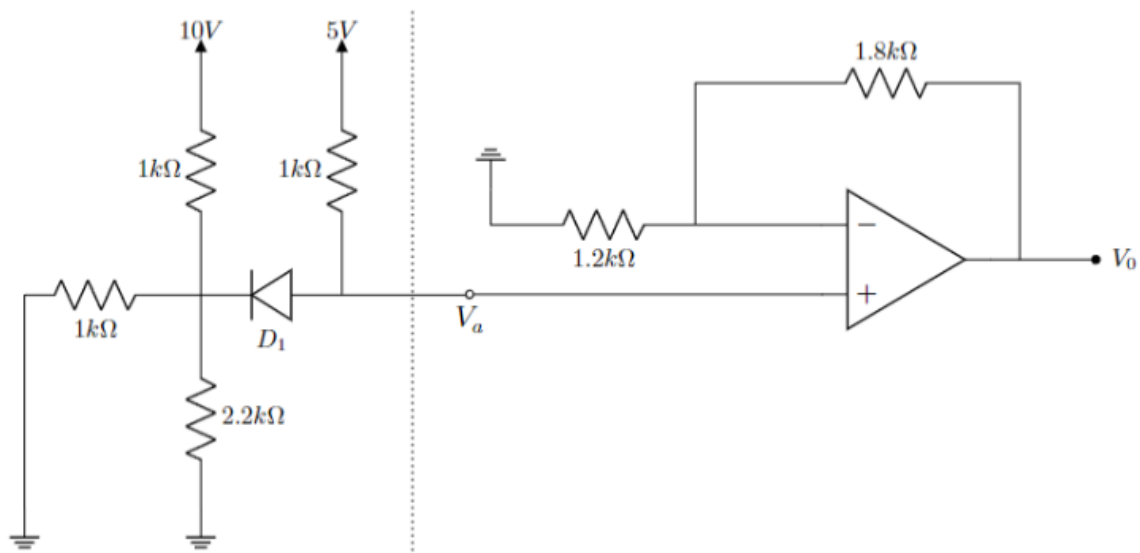


Figure 04

The saturation voltages of the diode are given as $V_s^+ = 10\text{V}$ and $V_s^- = -10\text{V}$. Assume $V_{D0} = 0.7\text{V}$

a) Determine the operating state of the diode. You must validate your assumption.

b) Calculate the value of V_a and V_o .

6. [CO1] **Draw** the I-V characteristics of a diode for Ideal, CVD and CVD+R model and show equivalent circuit for each model.